SCHOOL OF COMPUTING AND INFORMATICS | DEPARTMENT OF COMPUTER SCIENCE

Individual Assignment

CCS 3354: Data Mining — Weeks 1–6 | Total: 100 Marks

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Course Code:	CCS 3354
Course Title:	Data Mining
Submission Date:	4/3/2026

Declaration: I confirm that this assignment is entirely my own work. I have not copied from any other student or source without proper citation, and I understand the consequences of academic dishonesty.

Signature Date

SECTION A — Introduction to Data Mining & Data Warehousing [20 Marks]

Q1	**Knowledge Discovery in Databases (KDD) and Data Mining Fundamentals** Answer all four parts below.	10 marks

(a)	Define data mining and distinguish it from traditional database querying. In your answer, explain why data mining is necessary in modern data-driven organisations.	**(3 marks)**

Answer:

Data mining is the process of discovering meaningful patterns, relationships, trends, and knowledge from large volumes of data using statistical, machine learning, and computational techniques.

Difference Between Data Mining and Traditional Database Querying

Aspect	Data Mining	Traditional Database Querying
Purpose	Discover hidden patterns	Retrieve specific data
Approach	Exploratory & predictive	Exact & descriptive
Questions	“What patterns exist?”	“What is the value?”
Techniques	Machine learning, statistics	SQL queries
Output	Knowledge & models	Records & reports

In modern data-driven organisations, it is essential for informed decision-making, strategic planning, and maintaining competitive advantage in an environment characterized by massive and complex datasets.

(b)	Describe FOUR major functionalities of a data mining system, providing one real-world example for each functionality.	**(4 marks)**

Answer

1. **Concept Description (Characterization & Discrimination)**

Concept description summarizes general characteristics of a target class of data. It answers the question:What is typical of this group?"
There are two types:

a) **Characterization:** Describes features of a class.
b) Discrimination: Compares one class with another.

Real world example of Concept Description

A university analyzes student records to identify characteristics of students who graduate with First Class Honours

2. **Association Rule Mining**

Discovers relationships between variables in large datasets.

It answers: "Which items occur together?"

Uses measures like: support ,confidence and lift.

Real-World of Association rule mining

Supermarket market basket analysis:

If customers buy: bread and butter are also likely to buy Milk.

Retailers use this to arrange products strategically.

3. Classification and Prediction

Assigns data into predefined categories (classification) or predicts future values (prediction).It answers:"What class does this belong to?"

Real-World Example

Email spam detection:System classifies emails as spam or not spam

4. Cluster Analysis (Clustering)

Groups similar data objects together without predefined labels.

It answers: *"Which data items are similar?"*

Unlike classification, clustering is unsupervised.

Real-World Example

Telecommunication companies group customers based on usage patterns:Heavy data users,Frequent callers,Low-usage customers.This helps in targeted marketing and pricing plans.

(c)	Outline the six key steps of the KDD (Knowledge Discovery in Databases) process and briefly describe what happens at each step.	**(3 marks)**

Answer

The **KDD (Knowledge Discovery in Databases)** process is a systematic procedure used to extract useful knowledge from large datasets.

Steps:

1. Data Cleaning

This step removes noise, errors, and inconsistencies from the data to improve quality.

Activities: Handling missing values, Removing duplicates , Correcting, errors, Eliminating outliers

2. Data Integration

Combining data from multiple sources into a unified dataset.

Activities: Merging databases , Resoling naming conflicts , Removing redundancy

3. Data Selection

Selecting relevant data needed for the mining task.Not all data is useful.

Activities: Choosing relevant attributes , Filtering unnecessary records

4. Data Transformation

Converting data into suitable format for mining.

Activities: Normalization , Aggregation ,Feature construction , Discretization

5. Data Mining

Core step where algorithms are applied to extract patterns.

Techniques Used: Classification , Clustering, Association rules, Prediction

6. Pattern Evaluation and Knowledge Presentation

Identifying useful patterns and presenting them in understandable form.

Activities: Removing irrelevant patterns, Visualization, Reporting

Q2	**Data Warehousing Concepts and OLAP Operations** Answer all four parts below.	10 marks

(a)	Explain the fundamental architectural difference between an OLTP (Online Transaction Processing) system and an OLAP (Online	**(3 marks)**

	Analytical Processing) system. Use a concrete example to illustrate each.	

Answer

OLTP and OLAP are two different types of database systems designed for different purposes within an organisation. OLTP systems support daily operational transactions, while OLAP systems support complex data analysis and decision-making.

OLTP (Operational Systems)

Purpose: Handles daily transactions in real time.

Key Features:

- Fast inserts and updates
- Many users at once
- Stores current data
- Uses normalized tables

Example — ATM Withdrawal: When you withdraw money, the system checks your balance, deducts the amount, and updates your account instantly. This is OLTP.

OLAP (Analytical Systems)

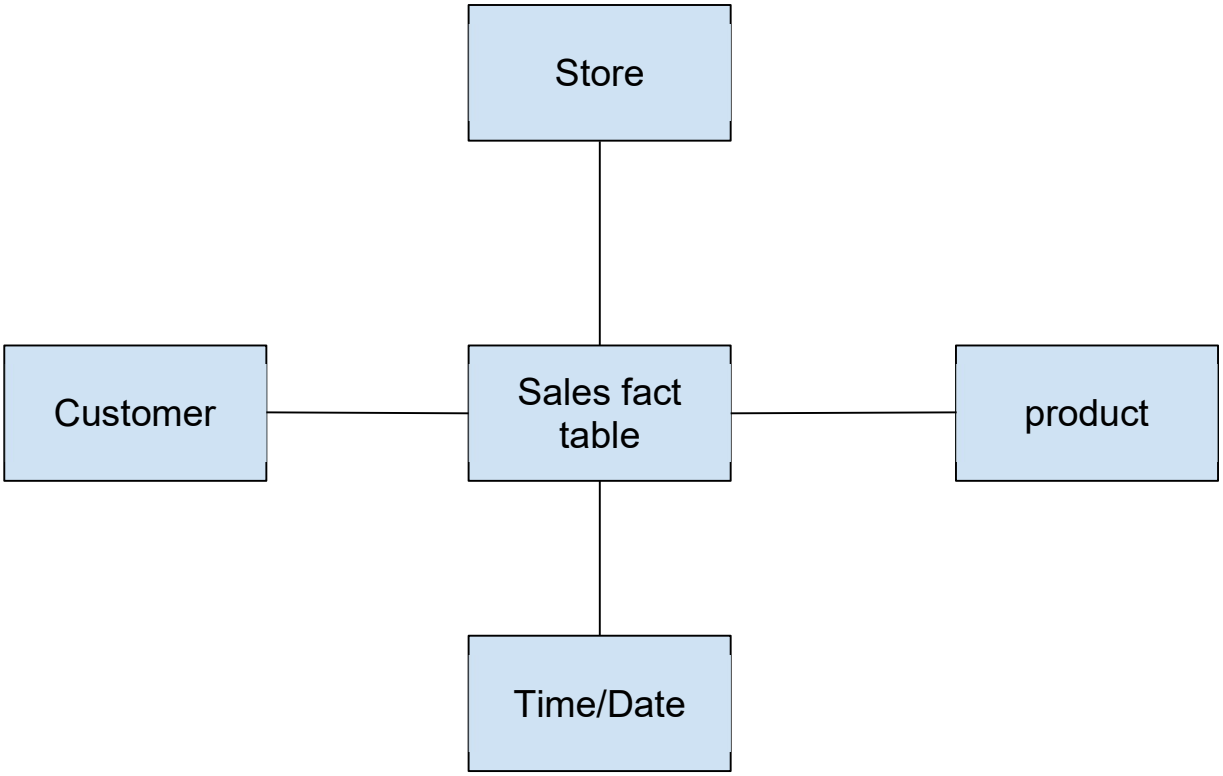
Purpose: Analyzes large amounts of historical data for decision-making.

Key Features:

- Complex queries
- Used by managers and analysts
- Stores historical data
- Uses data warehouses

Example — Bank Analysis: Managers analyze which branch made the most profit over the last 5 years using past transaction data. This is OLAP

(b)	A retail company has a data warehouse structured around sales transactions. Draw and label a Star Schema for this scenario, identifying the fact table and at least four dimension tables.	**(4 marks)**



Fact table : Sales

Dimension tables : Customer ,Product ,Date, Store.

(c)	Briefly explain THREE of the following OLAP cube operations, providing an example of each: (i) Roll-up, (ii) Drill-down, (iii) Slice, (iv) Dice.	**(3 marks)**

1. **Roll-up (Aggregation)**
   Roll-up summarizes detailed data into higher-level totals by moving up a hierarchy.**Example:**Sales data recorded per city (Nairobi, Mombasa) is aggregated to show total sales per country (Kenya).

2. Drill-down (Detailing)
Drill-down shows more detailed data by moving down a hierarchy (opposite of roll-up).**Example:**From yearly sales totals, an analyst drills down to view quarterly and monthly sales to find the best-performing month.

3. Slice

Slice selects one value of a dimension to create a smaller sub-cube.**Example:**
From a cube of Product × Region × Time, selecting Time = 2025 shows sales for all products and regions only in 2025.

SECTION B — Data Preprocessing: Parts 1 & 2 [25 Marks]

Q3	**Data Quality and Cleaning** Answer all three parts below.	10 marks

(a)	Identify and explain FOUR dimensions of data quality. For each dimension, provide an example of a problem that could arise and a corresponding remedy.	**(4 marks)**

Answer

1. Accuracy

Accuracy refers to how correctly data represents real-world values.

Problem Example:
A customer's age is recorded as 220 years due to a typing error.

Remedy:
Use validation rules (e.g., age must be between 0 and 120) and verification checks during data entry.

2. Completeness

Completeness means all required data fields are filled.

Problem Example:
Customer records missing phone numbers or addresses, making communication impossible.

Remedy:
Make important fields mandatory and use forms that prevent submission of incomplete records.

3. Consistency

Consistency ensures data is the same across different systems or datasets.

Problem Example:
A customer's address differs between the sales database and billing system.

Remedy:
Use a centralized database or synchronization processes to keep records uniform.

4. Timeliness

Explanation:
Timeliness refers to how up-to-date the data is.

Problem Example:
Using last year's inventory data to make current purchasing decisions.

Remedy:
Update data regularly and implement real-time data processing where necessary.

(b)	A student dataset contains the following age values: {22, 24, 21, 23, 150, 25, 22}. Identify the anomaly, name the technique you would use to detect it, and describe TWO methods for treating it.	**(3 marks)**

Identifying the Anomaly

The value **150** is the anomaly (outlier) because it is extremely different from the other ages, which range between 21 and 25. A student age of 150 years is unrealistic and likely caused by a data entry error.

Detection Technique

Outlier Detection using Statistical Methods (e.g., Z-score or Interquartile Range — IQR).

These methods identify values that are far from the majority of the data.

Example: Using IQR, any value far above the upper limit would be flagged as an outlier.

Methods for Treating the Outlier

1. Removal (Deletion)

Remove the incorrect value from the dataset if it is confirmed to be an error.

When Appropriate:
When the outlier is due to data entry mistakes or is impossible in reality.

2. Replacement (Imputation)
Replace the outlier with a more reasonable value, such as the mean or median of the dataset.**Example:**Replace 150 with the median age (23).

(c)	Distinguish between data integration and data transformation. Describe the entity identification problem that arises during data integration and explain how it is resolved.	**(3 marks)**

Data Integration

Data integration combines data from multiple sources into a single unified dataset.**Example:**Merging customer data from sales, billing, and support databases into one data warehouse.

Data Transformation

Data transformation converts data into a suitable format or structure for analysis.**Examples:**Normalizing values (e.g., converting marks to percentages)

Aggregating data (e.g., daily sales into monthly totals) ncoding categorical data into numerical form

Entity Identification Problem

The same real-world entity may appear differently in different datasets, making it hard to determine whether records refer to the same object.**Example:**One database records a student as “Simon Kamau,” while another records the same person as “S. Kamau.”

Resolution Methods

1. Use of Unique Identifiers

Assign a unique ID (e.g., student ID number) to each entity to ensure correct matching across datasets.

2. Data Matching Techniques

Use algorithms to compare attributes such as name, date of birth, and address to detect duplicates and merge records correctly.

Q4	**Data Transformation, Normalisation & Dimensionality Reduction** Answer all four parts below.	15 marks

(a)	The following dataset records students' exam scores: {45, 72, 88, 56, 91, 63, 77}. Apply Min-Max normalisation to map all values to the range [0, 1]. Show all working.	**(4 marks)**

Step 1: Identify Minimum and Maximum Values

Minimum (min) = 45
Maximum (max) = 91

Step 2: Use Min-Max Normalisation Formula

Normalized value = (x − min) / (max − min)

Range = 91 − 45 = 46

Step 3: Apply Formula to Each Value

45 → (45 − 45) / 46 = 0 / 46 = **0.000**
88 → (88 − 45) / 46 = 43 / 46 = **0.935**
91 → (91 − 45) / 46 = 46 / 46 = **1.000**
77 → (77 − 45) / 46 = 32 / 46 = **0.696**
72 → (72 − 45) / 46 = 27 / 46 = **0.587**
56 → (56 − 45) / 46 = 11 / 46 = **0.239**
63 → (63 − 45) / 46 = 18 / 46 = **0.391**

Final Normalized Dataset

{0.000, 0.587, 0.935, 0.239, 1.000, 0.391, 0.696}

(d)	Describe the five steps of Principal Component Analysis (PCA) and explain what is meant by 'variance explained' by a principal component. Under what circumstances might PCA be inappropriate?	**(3 marks)**

Step 1: Standardize the Data

Scale variables to have mean = 0 and standard deviation = 1 so that features with larger scales do not dominate the analysis.

Step 2: Compute the Covariance Matrix

Calculate how variables vary together to understand relationships between features.

Step 3: Compute Eigenvalues and Eigenvectors

Eigenvectors determine the directions of the principal components, while eigenvalues indicate the amount of variance captured by each component.

Step 4: Select Principal Components

Rank components by eigenvalues and keep the top components that capture most of the variance.

Step 5: Transform the Data

Project the original data onto the selected principal components to obtain a reduced-dimensional dataset.

Variance Explained

The variance explained by a principal component is the proportion of the dataset's total variability captured by that component. Components with higher explained variance contain more important information about the data.

When PCA May Be Inappropriate

- When data relationships are non-linear (PCA captures only linear patterns)
- When features are not numeric or cannot be meaningfully standardized
- When interpretability of original variables is required
- When variance does not represent importance (e.g., noise has high variance)

SECTION C — Data Mining Primitives & Query Languages [20 Marks]

Q5	**Data Mining Primitives and DMQL** Answer all three parts below.	10 marks

(a)	List and explain ALL FIVE data mining primitives, describing how each one contributes to specifying a complete data mining task. Use a cohesive example throughout (e.g., a banking scenario).	**(5 marks)**

1. Task-Relevant Data

Specifies the data to be used, including the database, tables, attributes, and conditions.**Bank Example:**Select customer records containing age, income, employment status, credit history, loan amount, and repayment status.

2. Kind of Knowledge to be Mined

Defines the type of patterns or knowledge to discover (e.g., classification, association, clustering, prediction).**Bank Example:**Use classification to predict whether a customer will default or repay a loan.

3. Background Knowledge

Incorporates domain knowledge such as concept hierarchies or constraints to guide the mining process.**Bank Example:**Income levels grouped as Low, Medium, High; age grouped as Youth, Adult, Senior.

4. Interestingness Measures Specifies criteria for evaluating the usefulness of discovered patterns (e.g., accuracy, support, confidence).**Bank Example:** Select rules with prediction accuracy above 85% for identifying high-risk customers.

5. Presentation of Discovered Patterns

Determines how results will be displayed to users (e.g., tables, charts, rules, decision trees).**Bank Example:**Display results as a decision tree showing factors that lead to loan default.

(b)	Write a valid DMQL query that mines association rules from a supermarket database called 'SuperMartDB'. Your query should reference a location hierarchy for customers, target the items and sales tables, and apply thresholds of min_support = 3% and min_confidence = 60%. Annotate each clause.	**(3 marks)**

```
MINE ASSOCIATION_RULES

ON SuperMartDB.Sales AS S

TARGET ITEMS(S.ItemID) FROM SuperMartDB.Customers AS C

HIERARCHY LOCATION(C.City, C.State, C.Country)

WHERE S.CustomerID = C.CustomerID

MIN_SUPPORT 0.03

MIN_CONFIDENCE 0.6
```

(c)	Explain the role of concept hierarchies in data mining. Provide two examples of concept hierarchies from different domains (e.g., time, geography, product category).	**(2 marks)**

Concept hierarchies (also called attribute hierarchies) are structures that organize data attributes into multiple levels of abstraction, from fine-grained (specific) to coarse-grained (general). They play a crucial role in knowledge discovery because they allow algorithms to analyze data at different levels, aggregate information, and identify patterns that might not be obvious at the raw data level.

1. **Data Generalization**
    - Concept hierarchies allow data to be **generalized** from detailed values to higher-level categories.
    - Example: Instead of analyzing purchases in every single city, you can aggregate by **state** or **country** to discover broader trends.
2. **Reducing Complexity**
    - By moving from fine to coarse levels, the algorithm handles **less detailed data**, reducing computational complexity.
3. **Pattern Discovery at Multiple Levels**

- Some patterns exist only at higher abstraction levels.
- Example: Frequent items may appear across categories rather than individual products.

4. **Facilitating Association Rule Mining**
   - Helps in discovering **generalized rules**, e.g., “Customers in *Kenya* buy *electronics*” instead of only city-specific patterns.
5. **Supports OLAP and Multidimensional Analysis**
   - Concept hierarchies are essential for **drill-down** and **roll-up** operations in multidimensional data analysis.

Examples of Concept Hierarchies

1. **Geography Hierarchy (Domain: Customer Location)**
   City → State → Country → Continent
   - **Fine-grained:** Nairobi, Mombasa
   - **Medium level:** Nairobi County, Mombasa County
   - **High level:** Kenya (Country), Africa (Continent)**Use:** Allows analysis of customer behavior at city, country, or continent level.
2. **Time Hierarchy (Domain: Sales or Transactions)**
   Day → Month → Quarter → Year
   - **Fine-grained:** March 3, 2026
   - **Medium level:** March 2026
   - **High level:** Q1 2026, 2026
   - **Use:** Enables trend analysis over months, quarters, or years instead of individual days.

Q6	**Data Mining System Architectures & Concept Description** Answer both parts below.	10 marks

(a)	Compare and contrast the FOUR levels of coupling between a data mining system and a database/data warehouse: No Coupling, Loose Coupling, Semi-Tight Coupling, and Tight Coupling. For each level, describe one practical advantage and one limitation. Present your answer in a structured manner.	**(6 marks)**

Answer:

1. No Coupling

The data mining system does not use any database or data warehouse functionality. Data is obtained from external sources such as files.

Advantage:
Simple to design and independent of database systems.

Limitation:
Poor performance and scalability because it cannot utilize database indexing, querying, or storage capabilities.

2. Loose Coupling

The data mining system uses the database only for data retrieval. All mining operations are performed outside the database.

Advantage:
Easy to implement using existing databases without major modifications.

Limitation:
Inefficient for large datasets because data must be transferred out of the database for processing.

3. Semi-Tight Coupling

The data mining system uses some database features such as indexing, sorting, aggregation, and query processing to improve performance.

Advantage:
Better efficiency since some operations are handled within the database system.

Limitation:
Still limited because core mining algorithms run outside the database.

4. Tight Coupling

The data mining system is fully integrated into the database or data warehouse. Mining functions are implemented as part of the database system.

Advantage:
High efficiency and scalability since data access and mining occur within the same system.

Limitation:
Complex to design and requires significant modification of the database system.

(b)	Distinguish between Data Characterisation and Data Discrimination (Comparison) as forms of concept description. Provide a clearly worked example of each using a dataset of your choice — the example must include the type of output produced (e.g., chart type, rule form).	**(4 marks)**

Data Characterisation

Summarizes the general features of a target class of data to describe what is typical about that class.

Example Dataset (University Students):
Attributes: Program, Average Score, Study Hours, Graduation StatusTarget Class: Graduated Students

Characterisation Result:
Most graduated students:

- Study more than 20 hours per week
- Have average scores above 65%
- Are mainly from Engineering and Computing programs

Type of Output Produced:

- Summary table
- Bar chart showing average scores by program
- Descriptive rules

Example Rule Form:
Graduated → StudyHours > 20 AND AvgScore > 65

Data Discrimination (Comparison)

Compares the target class with one or more contrasting classes to highlight differences between them.

Example Dataset (Same Students):
Compare Graduated vs Non-Graduated students

Discrimination Result:

Graduated Students:

- Higher study hours (20+ hours/week)
- Higher average scores (65%+)

Non-Graduated Students:

- Lower study hours (below 15 hours/week)
- Lower average scores (below 50%)

Type of Output Produced:

- Comparison table
- Side-by-side bar chart
- Classification rules

Example Rule Form:
IF StudyHours > 20 AND AvgScore > 65 THEN Graduated
ELSE Non-Graduated

SECTION D — Association Rule Mining: Part 1 [25 Marks]

Q7	**Support, Confidence, and Lift** Consider the following transaction database (N = 10 transactions): T1: {A, B, C} T2: {A, C} T3: {A, B, D} T4: {B, C, E} T5: {A, B, C, D} T6: {B, C} T7: {A, D} T8: {A, B, C} T9: {C, D, E} T10: {A, B} Use min_support = 50% and min_confidence = 70% for parts (a) to (c).	**10** marks
(a)	Calculate the support for each of the following itemsets: {A}, {B}, {C}, {A, B}, {A, C}, {B, C}, and {A, B, C}. Clearly show your working for each calculation.	**(4 marks)**

Support(X) = (Number of transactions containing X) / (Total transactions)

Total transactions N = 10

1. Support {A}

Transactions containing A: T1, T2, T3, T5, T7, T8, T10 → 7

Support(A) = 7 / 10 = 0.70 = 70%

2. Support {B}

Transactions containing B: T1, T3, T4, T5, T6, T8, T10 → 7

Support(B) = 7 / 10 = 0.70 = 70%

3. Support {C}

Transactions containing C: T1, T2, T4, T5, T6, T8, T9 → 7

Support(C) = 7 / 10 = 0.70 = 70%

4. Support {A, B}

Transactions containing A and B: T1, T3, T5, T8, T10 → 5

Support(A,B) = 5 / 10 = 0.50 = 50%

5. Support {A, C}

Transactions containing A and C: T1, T2, T5, T8 → 4

Support(A,C) = 4 / 10 = 0.40 = 40%

6. Support {B, C}

Transactions containing B and C: T1, T4, T5, T6, T8 → 5

Support(B,C) = 5 / 10 = 0.50 = 50%

7. Support {A, B, C}

Transactions containing A, B, and C: T1, T5, T8 → 3

Support(A,B,C) = 3 / 10 = 0.30 = 30%

{A} = 70%
{B} = 70%
{C} = 70%
{A,B} = 50%
{A,C} = 40%
{B,C} = 50%
{A,B,C} = 30%

Frequent Itemsets (min_support = 50%)

Frequent: {A}, {B}, {C}, {A,B}, {B,C}
Not Frequent: {A,C}, {A,B,C}

(b)	Determine which of the following rules are strong (satisfy both min_support and min_confidence): (i) {A} => {B}, (ii) {B} => {C}, (iii) {A, B} => {C}. Show all support and confidence calculations.	**(4 marks)**

Minimum support = 50%
Minimum confidence = 70%

Confidence Formula

Confidence ($X \Rightarrow Y$) = Support($X \cup Y$) / Support(X)

(i) Rule {A} ⇒ {B}

Support($A \cup B$) = Support(A,B) = 5/10 = 50%
Support(A) = 7/10 = 70%

Confidence = (5/10) ÷ (7/10) = 5/7 ≈ 0.714 = 71.4%

Decision:
Support = 50% (meets min_support)
Confidence = 71.4% (meets min_confidence)

Strong Rule

(ii) Rule {B} ⇒ {C}

Support($B \cup C$) = Support(B,C) = 5/10 = 50%
Support(B) = 7/10 = 70%

Confidence = (5/10) ÷ (7/10) = 5/7 ≈ 0.714 = 71.4%

Decision:
Support = 50%
Confidence = 71.4%

Strong Rule

(iii) Rule {A,B} ⇒ {C}

Support(A,B,C) = 3/10 = 30%
Support(A,B) = 5/10 = 50%

Confidence = (3/10) ÷ (5/10) = 3/5 = 0.60 = 60%

Decision:
Support = 30% (below min_support)
Confidence = 60% (below min_confidence)Not a Strong Rule

Final Answer Strong Rules:{A} ⇒ {B}{B} ⇒ {C}

Not Strong:{A,B} ⇒ {C}

(c)	Calculate the Lift for each rule identified as strong in part (b). Interpret each lift value — what does it reveal about the relationship between the items?	**(2 marks)**

Lift Formula

Lift (X ⇒ Y) = Confidence (X ⇒ Y) / Support(Y)

Strong Rules from Part (b):
{A} ⇒ {B}
{B} ⇒ {C}

(i) Rule {A} ⇒ {B}

Confidence = 0.714
Support(B) = 0.70

Lift = 0.714 / 0.70 ≈ 1.02

Interpretation:
Lift > 1 indicates a positive association. Customers who buy A are slightly more likely to also buy B compared to random chance, but the relationship is weak.

(ii) Rule {B} ⇒ {C}

Confidence = 0.714
Support(C) = 0.70

Lift = 0.714 / 0.70 ≈ 1.02

Interpretation:
Lift > 1 indicates a positive association. Customers who buy B are slightly more likely to also buy C, but the association is also weak.

Q8	**Apriori Algorithm & Market Basket Analysis** Answer all four parts below.	15 marks

(a)	State the Apriori property (anti-monotonicity principle) formally and informally. Explain why this property is critical to the efficiency of the Apriori algorithm and give a concrete example of how it enables candidate pruning.	**(4 marks)**

anti-monotonicity principle-If an itemset is frequent, then all of its non-empty subsets must also be frequent. Equivalently, if an itemset is infrequent, all of its supersets must also be infrequent.

Why This Property Is Critical to Apriori Efficiency

The Apriori algorithm uses this property to reduce the number of candidate itemsets it must check. Instead of testing all possible combinations, it prunes (eliminates) any candidate whose subset is already known to be infrequent, greatly reducing computation time.

Example (Market Basket — Supermarket)

Assume minimum support = 40%

Frequent 2-itemsets: {Milk, Bread}, {Bread, Butter}

Infrequent 2-itemset: {Milk, Butter}

When generating candidate 3-itemsets:

Possible candidate: {Milk, Bread, Butter}

Check its 2-item subsets:

- {Milk, Bread} → Frequent
- {Bread, Butter} → Frequent

- {Milk, Butter} → Infrequent

Since one subset is infrequent, the candidate {Milk, Bread, Butter} is pruned immediately without counting its support.

(b)	Using the transaction database from Question 7 with min_support = 50%, manually execute the Apriori algorithm. List: (i) all candidate 1-itemsets C1 and frequent 1-itemsets L1, (ii) all candidate 2-itemsets C2 (showing pruned candidates) and frequent 2-itemsets L2, (iii) all candidate 3-itemsets C3 and frequent 3-itemsets L3.	**(6 marks)**

(i) Candidate 1-itemsets C1 and Frequent 1-itemsets L1

C1 (all single items with support counts): A: 7, B: 7, C: 7, D: 5, E: 2

Support values: A = 70%, B = 70%, C = 70%, D = 50%, E = 20%

L1 (min_support = 50%): {A}, {B}, {C}, {D}

Item E is pruned (20% < 50%).

(ii) Candidate 2-itemsets C2 and Frequent 2-itemsets L2

Generate C2 from L1: {A,B}, {A,C}, {A,D}, {B,C}, {B,D}, {C,D}

Support counts: {A,B} = 5 → 50%
{A,C} = 4 → 40% (pruned)
{A,D} = 3 → 30% (pruned)
{B,C} = 5 → 50%
{B,D} = 1 → 10% (pruned)
{C,D} = 2 → 20% (pruned)

L2 (frequent 2-itemsets): {A,B}, {B,C}

(iii) Candidate 3-itemsets C3 and Frequent 3-itemsets L3

Generate C3 from L2: Possible join → {A,B,C}

Check subsets (Apriori property): {A,B} frequent
{B,C} frequent
{A,C} infrequent

Since subset {A,C} is infrequent, candidate {A,B,C} is pruned.

C3 = ∅ (no valid candidates) L3 = ∅ (no frequent 3-itemsets)

(c)	A large supermarket chain wants to use market basket analysis to improve sales. Describe THREE specific, actionable business decisions the supermarket could make based on discovered association rules. For each decision, explain which stakeholder would benefit and how.	**(3 marks)**

1. Product Placement Optimization

Decision: Place frequently co-purchased items (e.g., Bread and Butter) near each other in the store.

Stakeholder Benefiting: Customers and Store Management
Customers find related items easily, improving shopping convenience. Management benefits from increased impulse purchases and higher sales.

2. Cross-Selling Promotions and Bundles

Decision: Offer discounts or bundle deals on items often bought together (e.g., Buy Milk and get a discount on Cereal).

Stakeholder Benefiting: Marketing Team and Customers
The marketing team boosts sales of related products, while customers save money through promotions.

3. Inventory Planning and Stock Management

Decision: Ensure items that are frequently purchased together are stocked adequately at the same time.

Stakeholder Benefiting: Supply Chain Managers and Customers
Supply managers reduce stockouts and improve planning, while customers are more likely to find all needed items available.

SECTION E — Integrative Case Study [10 Marks]

Q9	**Case Study: National Health Monitoring Initiative** The Ministry of Health has commissioned a data mining project to analyse patient records from the past five years across 47 county hospitals. The dataset contains 2.3 million records with 120 attributes including: patient demographics, diagnoses (ICD-10 codes), lab results, medication prescriptions, length of stay, readmission rates, and geographic information. The data is sourced from disparate hospital information systems with varying formats, and is known to contain missing values, duplicate records, and inconsistent coding conventions. Answer all three parts below.	10 marks

(a)	Data Preparation Plan: As the lead data scientist, outline a comprehensive data preprocessing pipeline for this dataset. Your plan must address: (i) three specific data quality problems likely to exist in this dataset and how you would resolve each, (ii) which normalisation technique you would apply to numerical lab results and why, (iii) how you would reduce the 120-attribute dataset to the most relevant features before mining. Justify all choices.	**(4 marks)**

Data Preprocessing Pipeline

1. Data Cleaning (Resolving Data Quality Problems)

(i) Missing Values
Problem: Patient records may have missing lab results or incomplete demographic information.

Resolution: Use imputation methods such as mean/median substitution for numerical data and mode for categorical data, or remove records with excessive missing fields.

(ii) Duplicate Records
Problem: The same patient visit may appear multiple times due to integration from different hospital systems.
Resolution: Apply record linkage using unique patient IDs, timestamps, and demographic matching to detect and remove duplicates.

(iii) Inconsistent Coding Conventions
Problem: Diagnoses may be recorded using different ICD-10 formats or abbreviations across hospitals.
Resolution: Standardize codes using a unified ICD-10 reference and mapping tables.

2. Normalisation of Numerical Lab Results

Technique: Z-score normalization.

Reason: Lab results (e.g., blood pressure, glucose levels) are measured on different scales. Z-score normalization standardizes values based on mean and standard deviation, making them comparable without being affected by extreme values.

3. Feature Reduction (Reducing 120 Attributes)

(i) Remove Irrelevant and Redundant Attributes
Eliminate attributes that do not contribute to analysis (e.g., administrative IDs) and remove highly correlated features.

(ii) Principal Component Analysis (PCA)
Reduce dimensionality by transforming correlated variables into fewer components while preserving most of the information.

(iii) Attribute Selection Methods
Use statistical tests or machine learning methods to select features most strongly related to outcomes such as readmission rates.

(b)	Mining Strategy: The Ministry wants to discover which combinations of conditions, demographics, and medications are associated with 30-day hospital readmission. Specify a complete data mining task using the five primitives (task-relevant data, knowledge type,	**(4 marks)**

	background knowledge, interestingness measures, and presentation format). Then write a DMQL query implementing your specification.	

1. Task-Relevant Data

- **Tables:** Patients, Diagnoses, Medications, LabResults, Demographic
- **Attributes**
- **Patient demographics:** Age, gender, location (county/hospital), socioeconomic status
    - **Medical conditions:** ICD-10 diagnosis codes
    - **Medications:** Prescribed drugs and dosages
    - **Lab results:** Relevant lab values (normalized)
    - **Target:** Readmission30Days (Yes/No)
- **Filters:** Only include **admissions with complete records** and valid Readmission30Days values.

2. Knowledge Type

- **Association Rules**: Identify frequent co-occurrences of conditions, demographics, and medications linked to readmission.
  Example rule:
  {Age>65, Diabetes, Metformin} → {Readmission30Days = Yes}

3. Background Knowledge

- **Concept hierarchies:**
    - **Age:** Age → Age Group (0–18, 19–35, 36–60, 61+)
    - **Location:** City → County → Region
    - **Medications:** Drug → Drug Class
- **Domain knowledge:** Certain lab results or chronic conditions are known risk factors for readmission.

4. Interestingness Measures

- **Minimum Support:** 2% (patterns must occur in at least 2% of admissions)

- **Minimum Confidence:** 60% (rules must predict readmission correctly at least 60% of the time)
- **Lift / Leverage:** Optional, to highlight rules stronger than random chance.

5. Presentation Format

- **Tabular and ranked rules:**
  - Columns: Antecedent, Consequent, Support, Confidence, Lift
- **Hierarchical summary:**
  - Aggregate by location or age group for trend analysis
- **Graphical:**
  - Association network diagrams or heatmaps of conditions vs. readmission.

DMQL Query

```
MINE ASSOCIATION_RULES

ON HospitalDB.Admissions AS A

TARGET {A.DiagnosisCode, A.MedicationCode, A.AgeGroup, A.Gender}

FROM HospitalDB.Patients AS P

WHERE A.PatientID = P.PatientID

AND A.Readmission30Days IS NOT NULL

HIERARCHY AGE(P.Age, P.AgeGroup)

HIERARCHY LOCATION(P.City, P.County, P.Region)

MIN_SUPPORT 0.02

MIN_CONFIDENCE 0.6
```

(c)	Critical Evaluation: Identify TWO ethical concerns and ONE technical limitation specific to mining patient health records for association patterns. For each concern, propose a concrete mitigation strategy.	**(2 marks)**

1: Patient Privacy and Confidentiality

- Mining patient health records may expose sensitive personal information that could identify individuals.

- **Mitigation Strategy:**

    - Apply **data anonymization techniques**: remove direct identifiers (name, ID numbers) and mask quasi-identifiers (age, ZIP code) using generalization or k-anonymity.
    - Ensure compliance with **data protection regulations** (e.g., Kenya Data Protection Act 2019 or HIPAA equivalent).

2: Misuse or Discrimination

- Association patterns could be misinterpreted or misused to discriminate against patients based on health conditions, age, or location.

- **Mitigation Strategy:**
    - Limit access to mined patterns to **authorized medical professionals**.
    - Present results in **aggregated, non-identifiable form** (e.g., age groups, counties, not individuals).
    - Include **ethical review and oversight** before deploying patterns for decision-making.

Technical Limitation: Data Quality and Heterogeneity

- Patient records are from **multiple hospitals with varying formats, missing values, and inconsistent coding**, which can lead to biased or spurious association rules.

- **Mitigation Strategy:**

    - Implement a **robust data preprocessing pipeline**: standardize coding systems (ICD-10, LOINC), impute missing values, remove duplicates.

    - Conduct **data validation and quality checks** before mining.